

In addition to the online homework, please hand in the following problems:

For all of the following problems, assume C is a simple closed curve bounding a planar region R and oriented counter-clockwise.

Problem 1: Each pair of $\mathbf{F}$ and C below, compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

a. $\mathbf{F} = 2xy^3\mathbf{i} + 4x^2y^2\mathbf{j}$

C is the boundary of the a region in the first quadrant between: the x -axis, $x = 1$, and $y = x^3$.

b. $\mathbf{F} = [4x - 2y, 2x - 4y]$

C is the circle: $(x - 2)^2 + (y - 2)^2 = 4$.

Problem 2: Use Green's Theorem to verify the following identity:

$$\frac{1}{2} \oint_C -y dx + x dy = \text{Area of } R$$